Muhammad Sarab Ur Rehman

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Summary

Computer Engineering final-year student and applied AI/ML engineer with hands-on experience in Deep Learning, Generative AI (VAEs, GANs, Diffusion models, LLMs), Computer Vision, and NLP. Skilled across the full model lifecycle — data preprocessing, algorithm development, model training, explainability, and full-stack serving (FastAPI, React, Streamlit). Extending into AI hardware: designing and verifying a dataflow-reconfigurable CNN accelerator in SystemVerilog, with synthesis in Cadence Genus, as my final-year project. Completed AI/ML research and engineering internships at ITSOLERA, Ezitech, and Finetech, and hold Google AI and Data Analytics professional certificates.

Education

University of Engineering and Technology (UET), Lahore, Pakistan — Expected 2027
B.Sc. in Computer Engineering — *CGPA: 3.783 / 4.0*

Experience

AI & Generative AI Intern — Jun 2026 – Aug 2026
ITSOLERA Pvt Ltd — *Islamabad, Pakistan*

- Contributed to Artificial Intelligence and Generative AI projects, supporting model development, experimentation, and evaluation within a collaborative team.
- Delivered a physiologically grounded benchmarking suite evaluating synthetic EEG across 3 generative model families (VAE, GAN, Diffusion) as the core internship deliverable for Brain-Computer Interface research.

Machine Learning Intern — Mar 2026 – Jun 2026
Ezitech Institute — *Rawalpindi, Pakistan*

- Built **ClinIQ**, an AI clinical decision-support platform predicting disease risk across 6 conditions on public datasets ranging from 400 to 175K records, reaching up to 0.97–0.99 ROC-AUC.
- Delivered SHAP-based explainability and served predictions through a FastAPI backend and a React dashboard, with experiment tracking in MLflow.

Computer Vision Intern — Jan 2026 – Feb 2026
Finetech — *Chichawatni, Pakistan*

- Developed **GestureDesk**, a real-time touchless human-computer interaction system mapping webcam hand gestures to 12+ mouse, media, and OS-shortcut actions (cursor, click, scroll, volume, brightness, copy/paste, screenshot, alt-tab).
- Implemented 21-point hand-landmark tracking with MediaPipe and OpenCV, plus an optional face-recognition lock so gestures activate only for the enrolled user.

Projects

Dataflow-Reconfigurable CNN Accelerator | *Final Year Project* | *SystemVerilog, Python, Cadence Genus* — 2026

- Re-implementing a dataflow-reconfigurable (weight-/input-stationary) CNN accelerator from IEEE Micro (2026) in SystemVerilog, with 18 systolic PE arrays, collector units, and per-layer dataflow selection for Int8 edge inference.
- Building a parameterized design that generates 7 dataflow variants and verifying them bit-exactly against a NumPy reference on AlexNet, LeNet, VGG-16, and ConvNeXt layers, with area, power, and timing from Cadence Genus.

Synthetic Neural-Data Benchmarks for BCI | *GitHub* | *PyTorch, VAE, GAN, Diffusion, EEG* — 2026

- Built an open-source suite that evaluates AI-generated EEG using four neurophysiology-grounded metrics (spectral fidelity, phase–amplitude coupling, subject fingerprint, cross-session stability), replacing image-derived metrics such as FID.
- Trained 3 generative baselines (VAE, GAN, Diffusion) on the CHB-MIT epilepsy dataset and ranked them via a leaderboard report and interactive dashboard.
- Added a clinical safety protocol using a downstream classifier to flag synthetic data that is statistically plausible but clinically unsafe.

ClinIQ — AI Clinical Decision Support | *GitHub* | *XGBoost, SHAP, FastAPI, React, MLflow* — 2026

- Trained per-disease risk models across 6 conditions (diabetes, heart, stroke, kidney, liver, hypertension) on datasets from 400 to 175K records, with full data preprocessing, feature engineering, and training pipelines.
- Surfaced explainable predictions through a FastAPI service and a disease-selector React dashboard, with documented evaluation reports and data-quality limitations.

GestureDesk — Gesture-Controlled Desktop | *GitHub* | *MediaPipe, OpenCV, Python* — 2026

- Enabled full mouse, click, scroll, volume, brightness, and OS-shortcut control through real-time hand-gesture recognition, with adjustable sensitivity and toggleable gestures.
- Added an optional face-lock (enrollment + recognition) so gesture control responds only while the authorized user is in frame.

Technical Skills

Languages: Python, C/C++, SQL, R, JavaScript, LaTeX
Core ML / AI: Deep Learning, Machine Learning, Generative AI, Algorithm Development, Data Preprocessing, Feature Engineering, Model Evaluation
Frameworks: PyTorch, TensorFlow/Keras, Scikit-learn, XGBoost, LightGBM
Generative AI: VAEs, GANs, Diffusion Models, LLMs, Prompt Engineering, RAG
CV & NLP: OpenCV, MediaPipe, Speech-to-Text, Text-to-Speech, NLP
MLOps & Tools: FastAPI, Docker, MLflow, SHAP, Git/GitHub, React, Streamlit, Pandas, NumPy, Tableau

Certifications

Google AI — Professional Certificate, Google (Coursera), 2026

Google AI Essentials — Google (Coursera), 2025

Google Data Analytics — Professional Certificate, Google (Coursera), 2025

Google Project Management — Professional Certificate, Google (Coursera), 2025

Introduction to Git and GitHub — Google (Coursera), 2025

Workshops & Activities

Cyber Security Workshop — Cyber Security Society (CSS), UET Lahore

Participant, Scavenger Hunt — UTS 6.0, ACM UET Technical Summit